Yield, Duration, Convexity, and the Yield Curve

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LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Explain price sensitivity.** Use a fixed-income cash-flow model to establish the direction and curvature of its response to a yield change.
- **Compute local risk measures.** Calculate modified duration and convexity and use them to approximate a price change.
- **Read the yield curve.** Distinguish yield to maturity, par yield, and spot rates and explain which rates price individual cash flows.

1 A Safe Payment Can Have a Risky Price

L2a treated promised Treasury payments as known, but a holder who sells before maturity receives a market price. That price changes whenever the return available on comparable new securities changes. A fixed promise and a fluctuating resale value are therefore entirely compatible.

The 2023 failure of Silicon Valley Bank provides the motivating systems lesson. The bank combined long-duration fixed-rate assets, concentrated uninsured funding, and inadequate interest-rate and liquidity risk management. Rising rates reduced the fair value of its securities; deposit withdrawals then forced the institution to confront losses that a hold-to-maturity story had obscured. Bond sensitivity is thus one component of a coupled balance-sheet and liquidity problem, not a complete explanation by itself.

Burton Malkiel's bond-pricing principles organize the central comparative statics: price moves opposite yield, the price-yield curve is convex, longer maturity generally increases percentage sensitivity, and a larger coupon generally reduces it. These statements require an option-free fixed cash-flow model and comparisons that hold the other contract terms fixed.

2 Which Variables Move the Price?

Before restricting attention to the yield, it is worth asking which variables move a bond price at all. A zero-coupon Treasury bill answers that question cleanly, because a single promised payment gives a price that is smooth in both arguments. Let $V_P > 0$ dollars denote the face value paid at maturity, let $T > 0$ years denote the remaining time to maturity, let the positive integer n denote the number of compounding intervals per year, and let y denote the nominal annual yield with $1 + y/n > 0$. Holding n fixed, the bill price is $V_B(y,T) = V_P\left(1 + y/n\right)^{-nT}$, and its total differential is given by Eq. 1:

$$dV_B = \underbrace{\left(-V_P\left(1+y/n\right)^{-nT}\ \frac{T}{1+y/n}\right)}_{\partial V_B/\partial y} dy + \underbrace{\left(-V_P\left(1+y/n\right)^{-nT}\ n\log\left(1+y/n\right)\right)}_{\partial V_B/\partial T} dT. \quad (1)$$

The structure is the one a chemical engineer already knows from thermodynamic state functions: enumerate the independent variables, take one partial derivative per channel, and read off how each channel contributes. Here y is a market valuation variable while T is a contract term, so the two increments describe different experiments.

A third channel exists but is closed. Embedding the bill in the coupon-bearing family at a fixed payment count $N = nT$ gives $\partial V_B/\partial c = (V_P/n)\sum_{j=1}^{N}\mathcal{D}_{j,0}^{-1}(y;n) > 0$ for the annual coupon rate c, and this remains strictly positive when evaluated at $c = 0$. A bill therefore contributes no coupon term to Eq. 1 because its contract fixes $c = 0$, so $dc = 0$, and not because the coupon derivative vanishes.

Two cautions attach to the maturity channel. First, $\partial V_B/\partial T < 0$ requires $y > 0$: at $y = 0$ we have $\log(1 + y/n) = 0$ and maturity does not move the price, while for $-n < y < 0$ we have $\log(1 + y/n) < 0$ and the sign reverses. Second, dT conflates two experiments. Changing the contractual maturity compares two different securities, whereas the passage of holding time gives $dT = -dt$ along a single position. Both are smooth for a bill, whose price depends on T only through the exponent. Neither is globally smooth for a coupon security, whose payment count is an integer: between coupon events its dirty price varies smoothly with elapsed time, but a coupon leaving the schedule breaks that smoothness, a point taken up again below.

Finally, a total differential is linear in its increments by construction, so Eq. 1 enumerates the channels but cannot describe curvature within any one of them. The remainder of these notes does the complementary thing: fix the cash-flow schedule, keep only the yield channel, and go to second order.

3 Price as a Function of Yield

To isolate interest-rate risk, freeze the promised cash flows and vary only the yield used to discount them. The resulting yield-based valuation is defined locally in Defn. 3.1.

Definition 3.1. Yield-based bond price

Let the positive integer n denote the number of compounding intervals per year, let the positive integer N denote the number of remaining compounding periods, and let $X_j \geq 0$ dollars be the fixed cash flow paid at period index $j \in \{1, \dots, N\}$, with at least one $X_j > 0$. Let the nominal annual yield to maturity be y with $1 + y/n > 0$. Under a flat-yield convention, the time-0 dirty price $V_B(y)$, meaning the value including any accrued interest, is given by Eq. 2:

$$V_B(y) := \sum_{j=1}^{N} X_j \left(1 + \frac{y}{n}\right)^{-j}. \tag{2}$$

The yield y is the single internal discount rate applied to every promised payment; it is not a complete term structure of spot rates. Permitting $X_j = 0$ on periods carrying no payment lets a zero-coupon bill share this definition, with $X_N = V_P$ its only nonzero flow. The integer exponents place the valuation date on a coupon date; a general settlement date requires fractional-period timing conventions that these notes do not need.

The j-period discount factor $\mathcal{D}_{j,0}^{-1}(y;n) = (1 + y/n)^{-j}$ converts the notation in Eq. 2 into a sum of nonnegative discounted claims, at least one of which is strictly positive. Differentiating that sum establishes both the direction and curvature of the price response (Thm. 3.1).

Theorem 3.1. Monotonicity and convexity of an option-free bond

Under the assumptions of Defn. 3.1, assume $1 + y/n > 0$. The first and second derivatives of the dirty price with respect to the nominal annual yield are given by Eq. 3:

$$\frac{dV_B}{dy} = -\frac{1}{n}\sum_{j=1}^{N} jX_j\,(1+y/n)^{-j-1} < 0, \qquad \frac{d^2V_B}{dy^2} = \frac{1}{n^2}\sum_{j=1}^{N} j(j+1)X_j\,(1+y/n)^{-j-2} > 0. \tag{3}$$

The strict inequalities hold because no term of either sum is negative and at least one term of each is positive. Consequently, the bond price is strictly decreasing and strictly convex as a function of yield. The result assumes a fixed, nonnegative, and not identically zero cash-flow schedule, and therefore does not automatically apply to securities whose cash flows change when rates change, such as callable bonds.

The signs in Eq. 3 explain the price-yield geometry: a higher yield lowers every discounted payment, while positive curvature makes the gain from a yield decrease larger than the loss from an equal yield increase, measured from the same starting yield (Fig. 1).

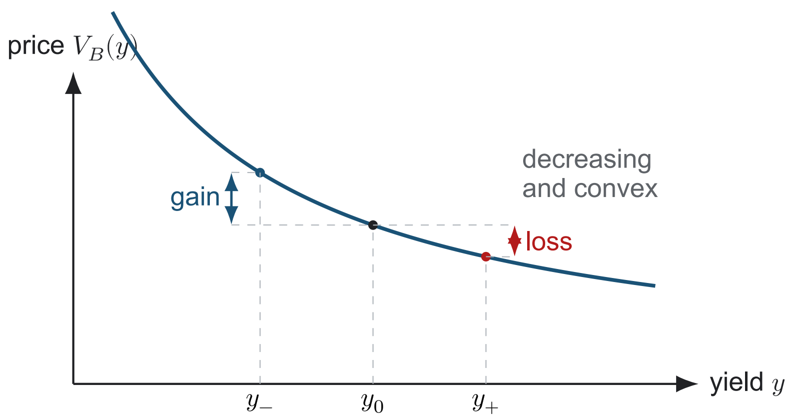

Figure 1: The price-yield curve for a fixed positive cash-flow stream. The comparison yields are $y_- := y_0 - \Delta y$ and $y_+ := y_0 + \Delta y$ for an equal change $\Delta y > 0$ from the initial yield y_0. Convexity makes the vertical price gain from the rate decrease exceed the price loss from the equal rate increase.

The bill-sensitivity notebook tests the first part of Thm. 3.1 using zero-coupon Treasury auction data, where the single maturity payment makes the mechanism especially transparent.

COMPANION NOTEBOOK
L2b: Treasury Bill Price Sensitivity →
Measure how a zero-coupon bill price changes with yield and remaining time using Treasury auction records.

4 Duration and Convexity

The derivative signs answer *which direction*; risk management also needs *how much*. Normalizing the derivatives by price produces two local measures with interpretable units (Defn. 4.1).

Definition 4.1. Modified duration and convexity

For the fixed cash-flow schedule, nominal annual yield y, coupon frequency n, and price $V_B(y) > 0$ defined in Defn. 3.1, the modified duration $D_{\text{mod}}(y)$ and convexity $K(y)$ are defined by Eq. 4:

$$D_{\text{mod}}(y) := -\frac{1}{V_B(y)}\frac{dV_B}{dy}, \qquad K(y) := \frac{1}{V_B(y)}\frac{d^2V_B}{dy^2}. \tag{4}$$

When the yield y is expressed as a decimal per year, modified duration has units of years and convexity has units of years squared. Both measures are positive under the assumptions of Thm. 3.1.

For a small yield shock Δy, a second-order Taylor approximation combines the two normalized sensitivities. The approximate fractional price change is given by Eq. 5:

$$\frac{\Delta V_B}{V_B} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}K(\Delta y)^2. \tag{5}$$

The duration term supplies the local slope, while the convexity term corrects the straight-line approximation for curvature. This formula uses a decimal yield change: a 25-basis-point shock is $\Delta y = 0.0025$, where one basis point is 10^{-4}.

5 How Contract Structure Changes Sensitivity

The *Macaulay duration* D_{mac} is the present-value-weighted average time to payment, and the modified duration of Defn. 4.1 is that average discounted by one period, $D_{\text{mod}} = D_{\text{mac}}/(1 + y/n)$. Because the two differ only by that positive factor, both respond to payment timing in the same way: moving more value toward distant dates increases exposure to discount-rate changes, and moving value toward earlier dates reduces it. That principle gives the carefully qualified maturity and coupon comparisons associated with Malkiel.

Maturity effect. Assume $y > 0$ throughout this comparison. Malkiel's maturity statements compare bonds selling at par, and that restriction supplies a closed form. Summing the geometric series in Eq. 2 for a coupon rate c gives $V_B/V_P = c/y + (1 - c/y)\,(1 + y/n)^{-N}$. Differentiating with respect to y at fixed c, and only then setting $c = y$ so that $V_B = V_P$, yields the par-bond modified duration and its increment, given by Eq. 6:

$$D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}, \qquad D_{\text{mod}}(N+1) - D_{\text{mod}}(N) = \frac{(1 + y/n)^{-N-1}}{n}. \tag{6}$$

The first expression increases strictly with N and is bounded above by $1/y$, the modified duration of a perpetuity, while the second shows the successive increments shrinking by the constant factor $(1 + y/n)^{-1} < 1$. Those two facts are Malkiel's maturity results. Because a par bond has the same initial price V_P at every maturity, the dollar sensitivity $-dV_B/dy = V_P D_{\text{mod}}(N)$ inherits both properties, so the percentage and dollar rankings agree here. Outside the par comparison they need not: for a zero-coupon security, modified duration grows without bound in T, yet the dollar sensitivity $-dV_B/dy = V_P T\,(1 + y/n)^{-nT-1}$ rises, peaks at $T = 1/\left(n \log\left(1 + y/n\right)\right)$, and then declines, because the shrinking price eventually outweighs the growing duration. These are all comparisons of yield sensitivity, not a claim that adding another coupon period always lowers the bond's price.

Coupon effect. Holding yield, maturity, face value, and coupon frequency fixed, a higher coupon shifts a larger fraction of present value toward earlier dates. The resulting percentage price sensitivity is therefore lower. The comparison requires at least two remaining payment dates: if a single payment remains, all value arrives on one date and the modified duration is $1/(n(1 + y/n))$ years regardless of the coupon.

Discrete-maturity caution. A coupon count N is an integer. Extending maturity adds a payment and changes the cash-flow vector, so differentiating a finite coupon sum with respect to a continuous maturity variable can be misleading. Compare two explicitly defined cash-flow streams or use duration with the cash flows held fixed.

6 From One Yield to a Curve of Discount Rates

Yield to maturity compresses a security's price and complete promised cash-flow schedule into one internal rate of return. That scalar is useful for quoting and comparison, but it is not the collection of discount rates used by an arbitrage-consistent term-structure model. A *spot rate* applies to a single payment at one maturity; the resulting spot curve supplies one discount factor for each payment date. A *par yield* is the coupon rate that makes a hypothetical coupon security trade at par under that discount curve.

For cash flows X_j paid at times t_j, a term-structure valuation replaces the flat-yield expression in Eq. 2 with

$$V_B = \sum_{j=1}^{N} X_j D(0, t_j), \tag{7}$$

where $D(0, t_j)$ is the time-0 discount factor for maturity t_j. The curve's level, slope, and curvature summarize how required compensation varies across maturities; they do not by themselves identify a unique forecast of future short rates. This distinction is the essential bridge from fixed income to the multi-period asset models used in Week 3.

The coupon-security notebook evaluates these comparative statics on Treasury note data and shows how coupon placement changes the response to a yield shock.

COMPANION NOTEBOOK

L2b: Treasury Duration and Convexity →

Price a coupon Treasury security and compare its yield, coupon, duration, and convexity sensitivities.

Optional advanced paths

Students who want the full theorem proofs, empirical curve exercises, stochastic short-rate models, or STRIPS construction can continue in `week-2/L2b/advanced/`. These extensions are deliberately separated from the required sequence so that Week 3 can begin with equity markets.

KEY TAKEAWAYS

In this lecture, we differentiated a fixed cash-flow bond price, established its decreasing convex price-yield shape, and converted those derivatives into duration and convexity risk measures.

- **Yield and price oppose.** For a fixed positive cash-flow stream, a higher yield lowers price because every promised payment is discounted more heavily.
- **Curvature creates asymmetry.** Positive convexity makes the gain from a yield decrease larger than the loss from an equal yield increase around the same starting yield.
- **A yield is not a curve.** Payment timing controls duration; yield to maturity is a single internal rate, while spot discount factors price individual cash flows and par yields describe hypothetical securities priced at par.

Next, carry these return, risk, and time-scale ideas into equity markets and empirical growth-rate data.

ADDITIONAL READING

- Burton G. Malkiel, "Expectations, Bond Prices, and the Term Structure of Interest Rates," *Quarterly Journal of Economics* 76(2), 197–218 (1962).
- Federal Reserve Board, *Review of the Federal Reserve's Supervision and Regulation of Silicon Valley Bank* (2023), for the interaction of interest-rate, liquidity, and governance risk.
- Frank J. Fabozzi and Francesco A. Fabozzi, *Bond Markets, Analysis, and Strategies*, 10th ed. (2021), for duration and convexity conventions.